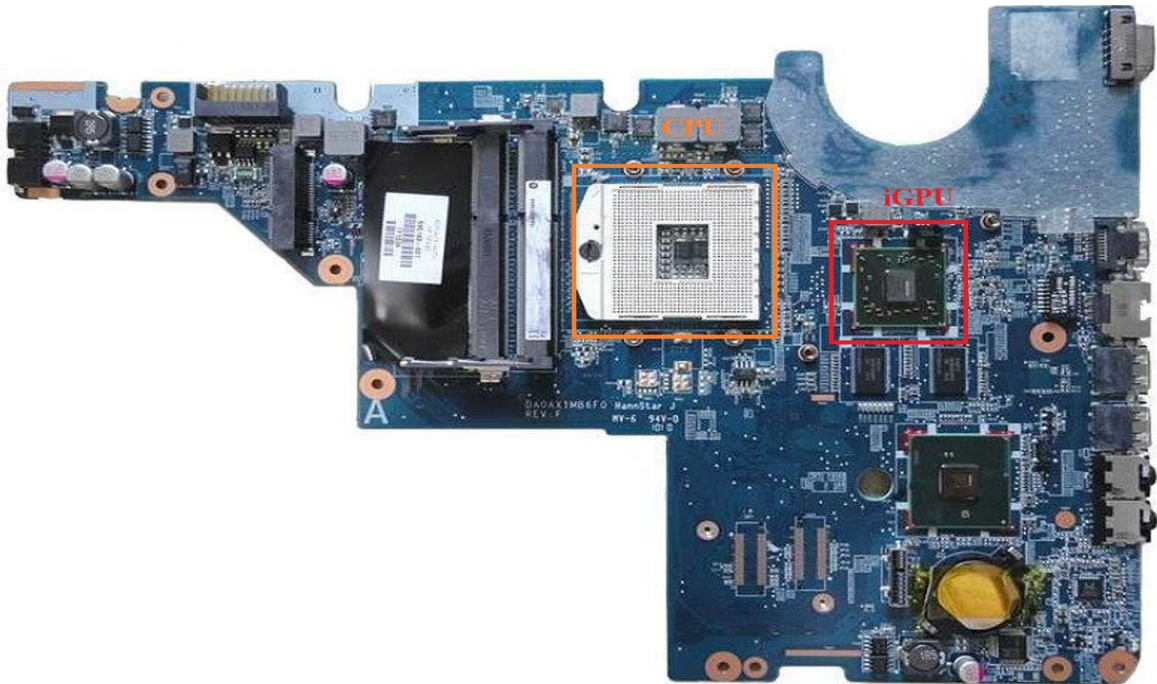


HARDWARE

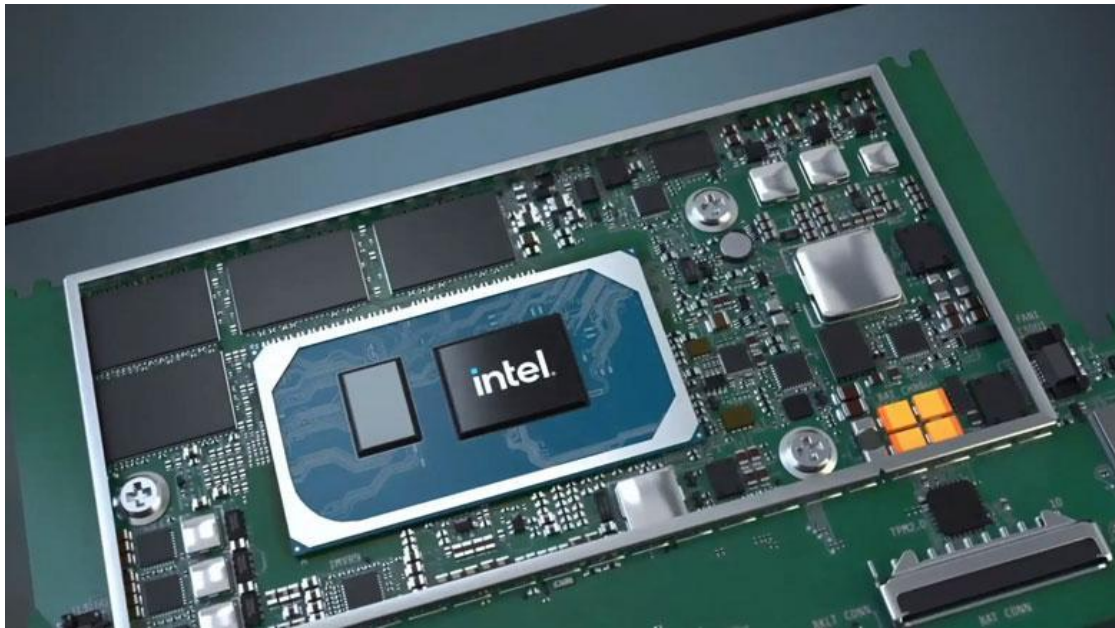
TASK 1:

Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.



I examined the graphics hardware of an HP laptop with an Intel Core i3 processor. The GPU installed in this system is **Intel UHD Graphics**. The brand of the GPU is **Intel**, and since it comes with an Intel Core i3 processor, the exact model may vary slightly depending on the generation (such as Intel UHD Graphics 620, 630, or similar).

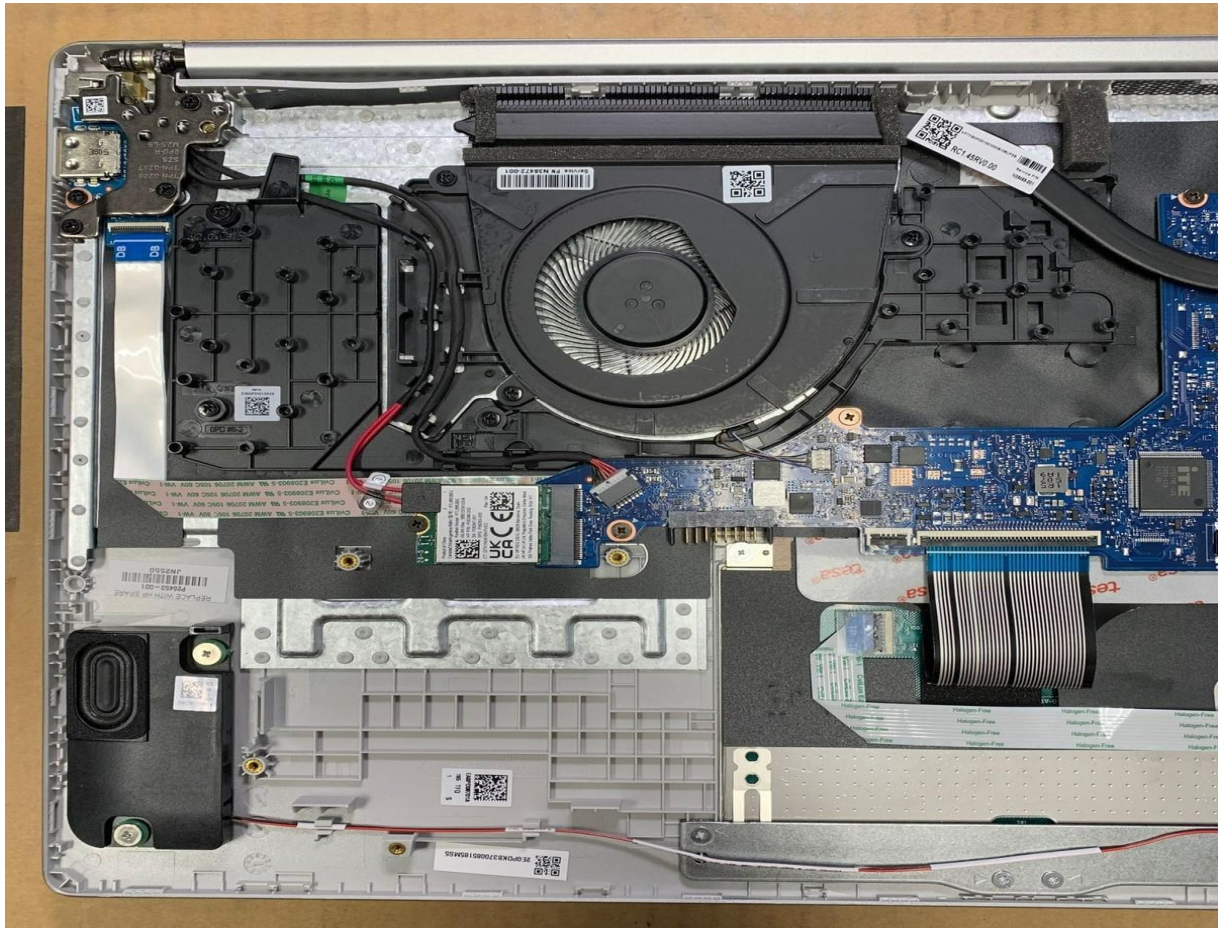
This GPU is an **integrated graphics processor**, which means it is built directly into the CPU and does not exist as a separate physical component inside the laptop. It uses a portion of the system's RAM for graphics processing instead of having its own dedicated memory. Integrated GPUs like Intel UHD Graphics are commonly found in laptops because they consume less power, generate less heat, and are suitable for everyday tasks such as web browsing, watching videos, using applications, and light gaming.



- **Brand:** Intel
- **Model:** Intel UHD Graphics (exact model depends on generation)
- **Type:** Integrated GPU

TASK 2:

Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

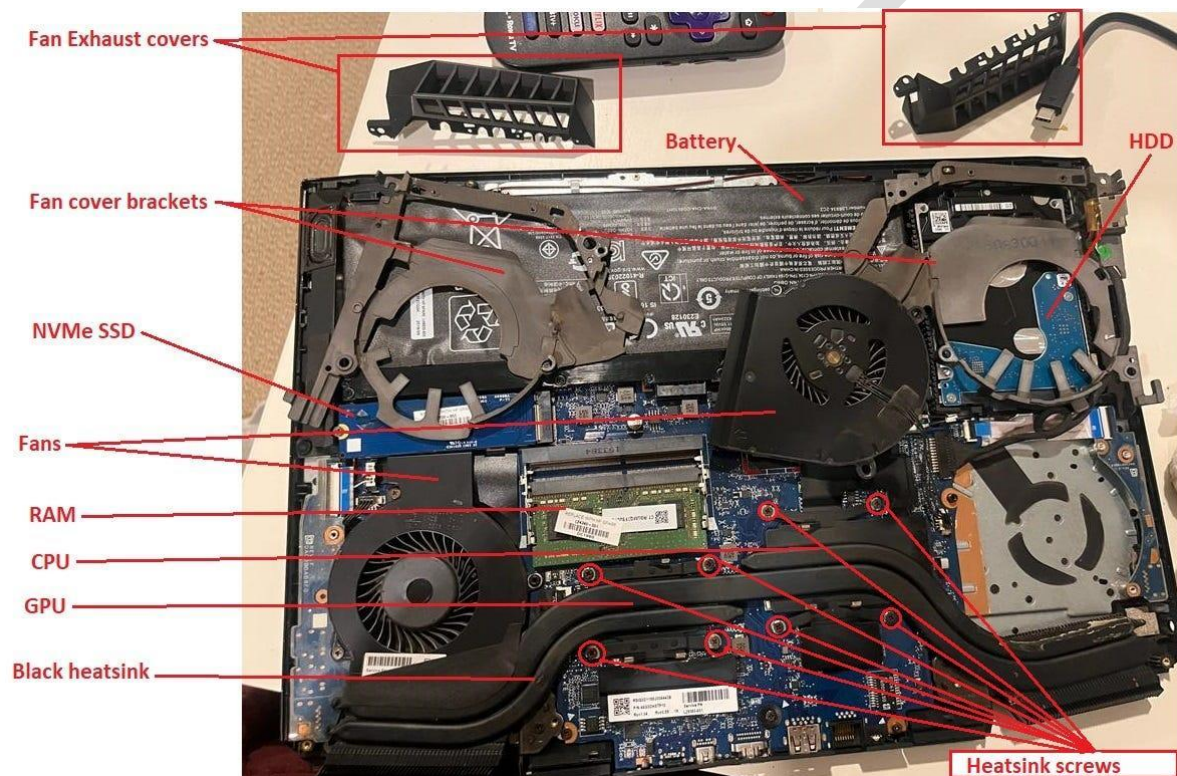


The main components of the cooling system are the **cooling fan, heatsink, and heat pipe**. The **cooling fan** is a small, high-speed fan that draws cool air into the laptop and pushes hot air out through the vents. It plays a crucial role in maintaining airflow inside the system. The **heatsink** is usually made of metal (aluminum or copper) and is attached to the CPU. Its function is to absorb the heat generated by the processor and spread it across a larger surface area for easier cooling. The **heat pipe** is a sealed copper tube that connects the CPU to the heatsink. It transfers heat efficiently using a liquid evaporation and condensation process inside the pipe.

When the laptop is running, the CPU generates heat. This heat is transferred through the heat pipe to the heatsink. The cooling fan then blows air over the heatsink, which removes the heat and expels it outside the laptop. This process ensures that the system remains cool, stable, and performs efficiently even during continuous use.

Therefore, the labeled components and their functions are:

- **Cooling Fan** – Circulates air and removes hot air from the system
- **Heatsink** – Absorbs and dissipates heat from the CPU
- **Heat Pipe** – Transfers heat from CPU to heatsink efficiently



TASK 3:

Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling System	Liquid Cooling System
Cost	Generally cheaper and more affordable	More expensive due to advanced components
Cooling Efficiency	Good for normal usage and moderate workloads	Highly efficient; better for heavy tasks and gaming
Maintenance	Low maintenance; occasional cleaning required	Requires regular maintenance and monitoring of liquid
Installation	Easy to install and widely available	More complex installation process
Noise Level	Can be noisy due to fans	Usually quieter as it uses liquid flow instead of fast fans
Size/Space	Takes up more space due to large fans and heatsinks	More compact in some setups but includes radiator

TASK 4:

Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

If I were building a gaming PC for streaming IPL matches and playing graphics-heavy games, I would choose **liquid cooling for the GPU**. Liquid cooling provides better heat dissipation, which is important during long gaming and streaming sessions where the GPU works continuously at high performance. It also helps maintain lower temperatures and quieter operation compared to air cooling. Although it is more expensive and requires more maintenance, the improved efficiency and performance make it a better choice for a high-end gaming and streaming setup.